

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.2283194895068398, median 0.2093242238725194, std: 0.13027792693766801
Reprojection error (cam1): mean 0.24616453939562097, median 0.22492542928712936, std: 0.1411821281886932
Gyroscope error (imu0): mean 0.35333926592297066, median 0.2902888191493197, std: 0.26522779364801763
Accelerometer error (imu0): mean 0.8341096054350354, median 0.8139536775959468, std: 0.273189095685587

Residuals

Reprojection error (cam0) [px]: mean 0.2283194895068398, median 0.2093242238725194, std: 0.13027792693766801
Reprojection error (cam1) [px]: mean 0.24616453939562097, median 0.22492542928712936, std: 0.1411821281886932
Gyroscope error (imu0) [rad/s]: mean 0.007900906177236374, median 0.006491055327260216, std: 0.005930673761192544
Accelerometer error (imu0) [m/s^2]: mean 0.1865125778438267, median 0.1820055753640485, std: 0.061086938866466704

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.00458484  0.99980452 -0.01923298  0.07506412]
 [ 0.99995614 -0.00442676  0.00825391  0.00046685]
 [ 0.00816716 -0.01926998 -0.99978096 -0.01556867]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.00458484  0.99995614  0.00816716 -0.00068384]
 [ 0.99980452 -0.00442676 -0.01926998 -0.07534739]
 [ -0.01923298  0.00825391 -0.99978096 -0.01412541]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
0.002845417633379418

Transformation (cam1):

T_ci: (imu0 to cam1):

```
[[ 0.00102972  0.99995801 -0.00910648 -0.00086081]
 [ 0.99998574 -0.00098194  0.00525011  0.00209483]
 [ 0.00524095 -0.00911175 -0.99994475 -0.01563484]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam1 to imu0):

```
[[ 0.00102972  0.99998574  0.00524095 -0.00201197]
 [ 0.99995801 -0.00098194 -0.00911175  0.00072037]
 [-0.00910648  0.00525011 -0.99994475 -0.01565281]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.0029465236788759246

Baselines:

Baseline (cam0 to cam1):

```
[[ 0.9999424 -0.00347206 -0.01015628 -0.07607711]
 [ 0.00350205 0.99998955  0.002937  0.00141083]
 [ 0.01014598 -0.0029724  0.99994411 -0.00082725]
 [ 0.         0.         0.         1.         ]]
```

baseline norm: 0.07609468702492003 [m]

Gravity vector in target coords: [m/s^2]
[0.06044797 -9.80436468 0.1979955]

Calibration configuration

=====

cam0

Camera model: pinhole
Focal length: [400.4814123024703, 400.6352560522375]
Principal point: [322.33123328330237, 205.48166803116757]
Distortion model: radtan
Distortion coefficients: [0.03318323418584644, -0.10790505791794129, 0.0010976800751699432, -0.0019812875905540167]
Type: aprilgrid
Tags:
Rows: 5
Cols: 5
Size: 0.04 [m]
Spacing 0.012 [m]

cam1

Camera model: pinhole
Focal length: [399.087067623455, 399.12724497515325]
Principal point: [330.7662134787063, 201.01493203166558]
Distortion model: radtan
Distortion coefficients: [0.015125799257507401, -0.05698162976293714, -0.002997141930509312, 0.0015872773728161814]
Type: aprilgrid
Tags:
Rows: 5
Cols: 5
Size: 0.04 [m]
Spacing 0.012 [m]

IMU configuration

=====

IMU0:

Model: calibrated
Update rate: 125.0

Accelerometer:

Noise density: 0.02

Noise density (discrete): 0.22360679774997896

Random walk: 0.0002

Gyroscope:

Noise density: 0.002

Noise density (discrete): 0.022360679774997897

Random walk: 2e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

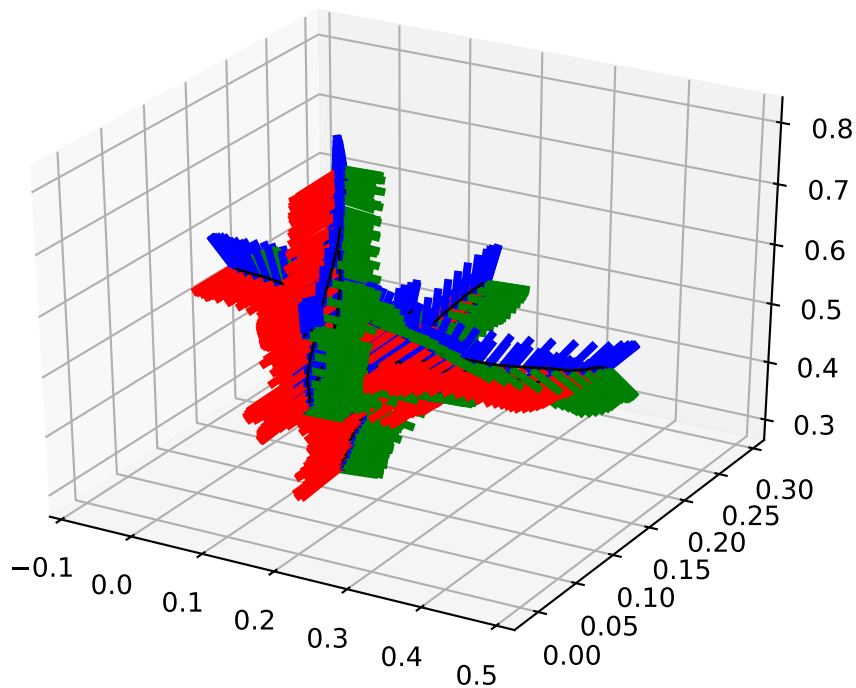
[0. 1. 0. 0.]

[0. 0. 1. 0.]

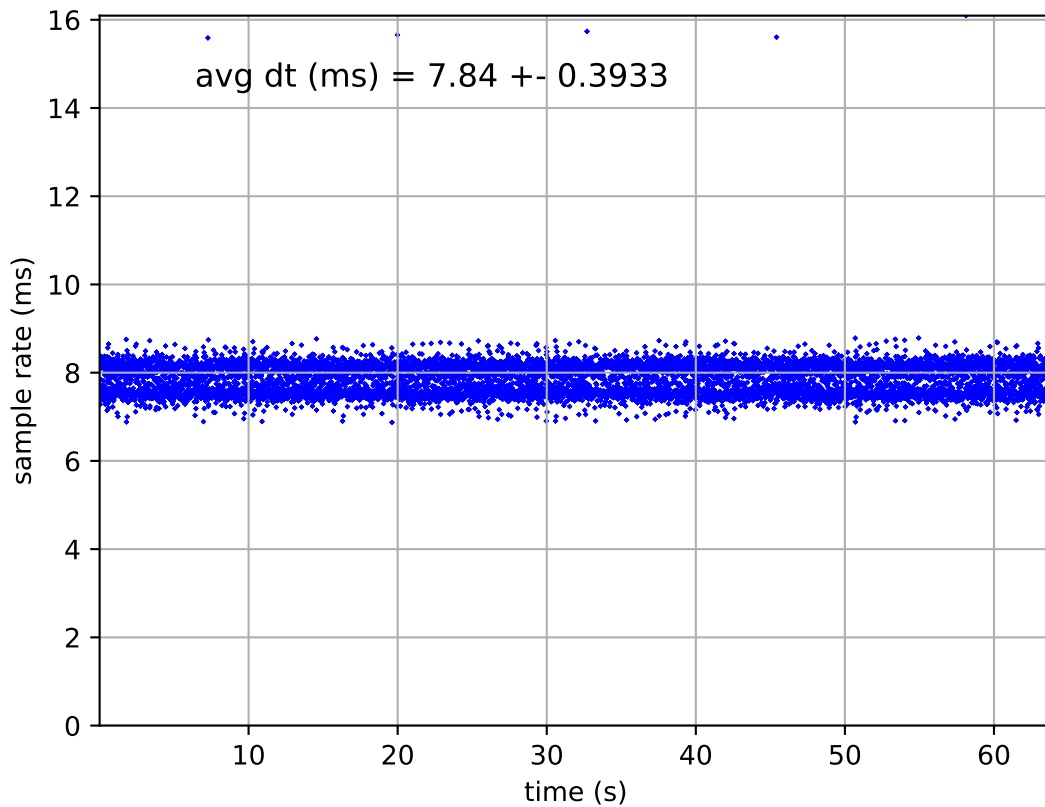
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

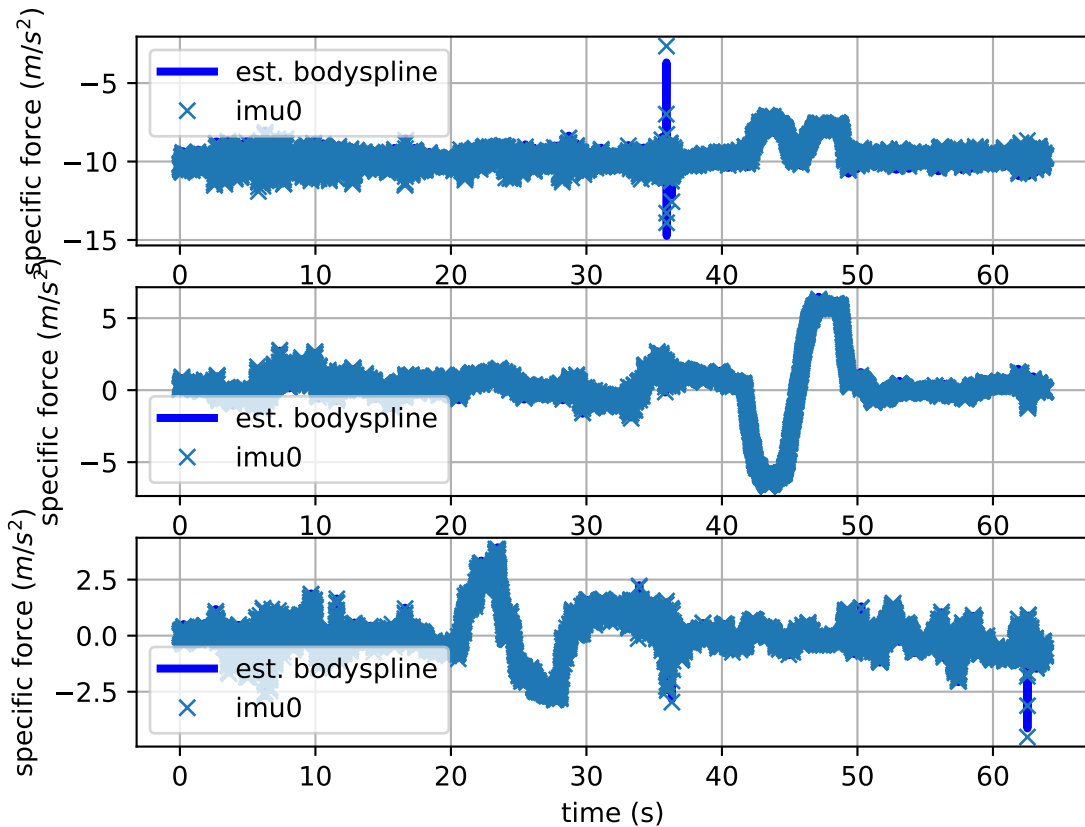
imu0: estimated poses



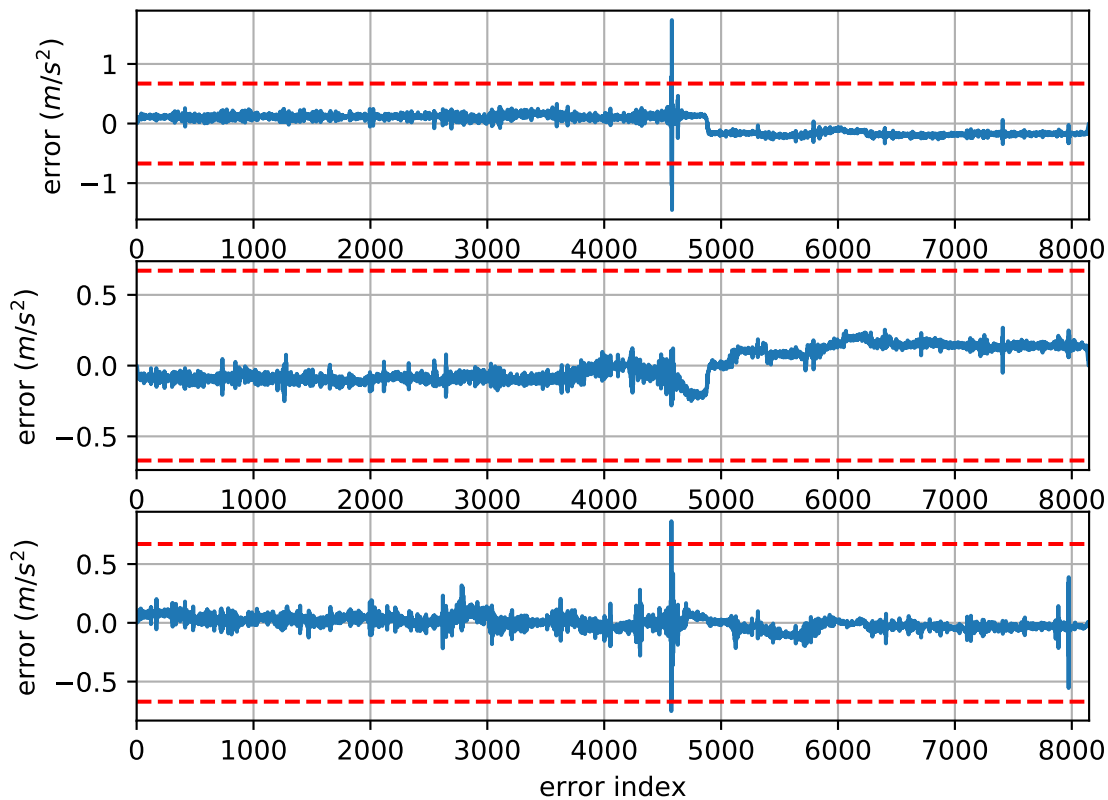
imu0: sample inertial rate



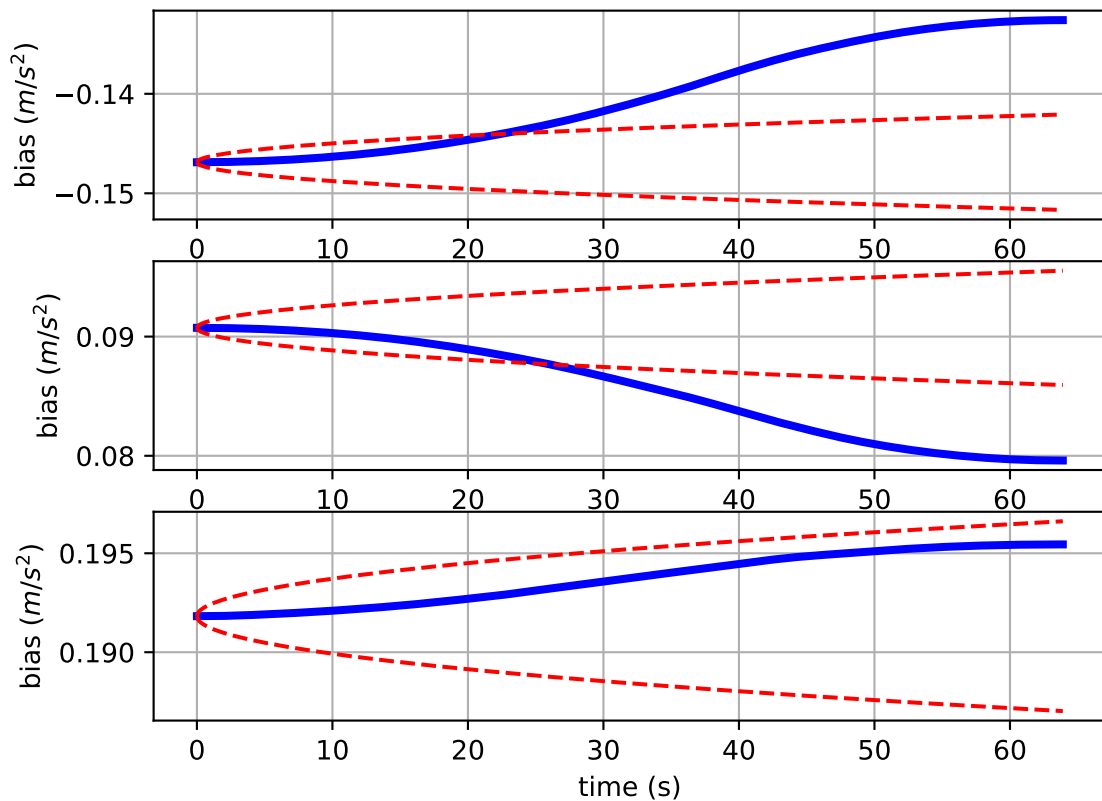
Comparison of predicted and measured specific force (imu0 frame)



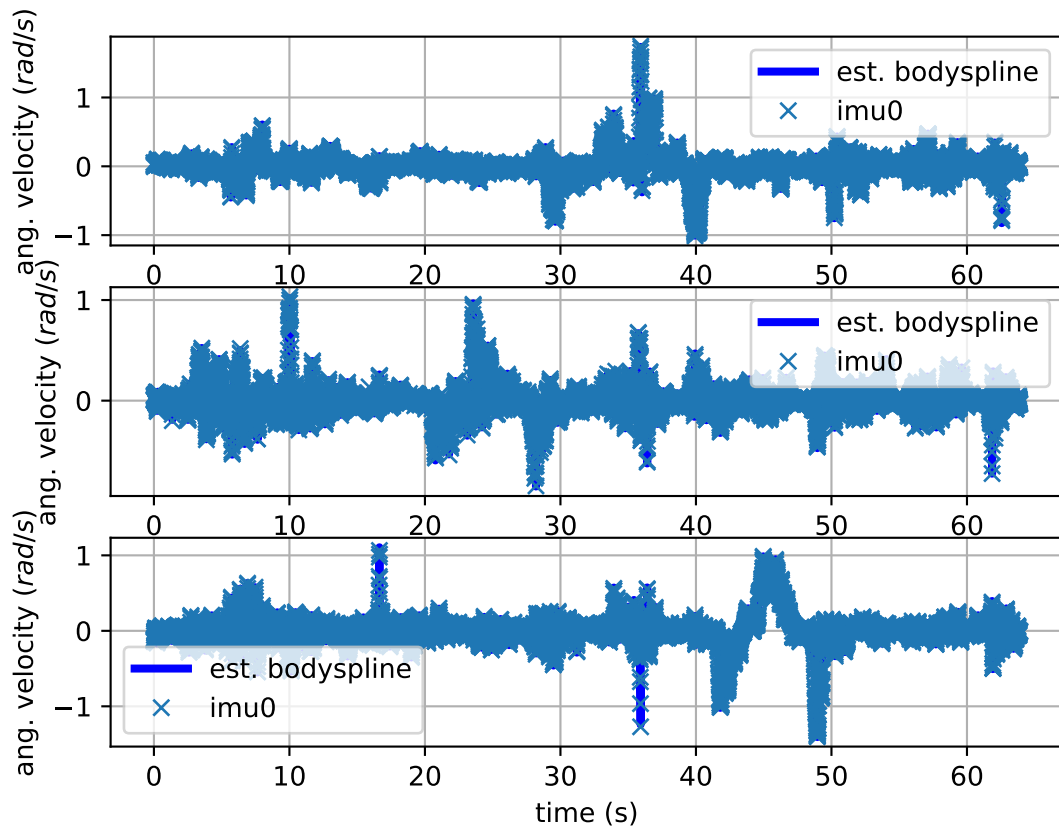
imu0: acceleration error



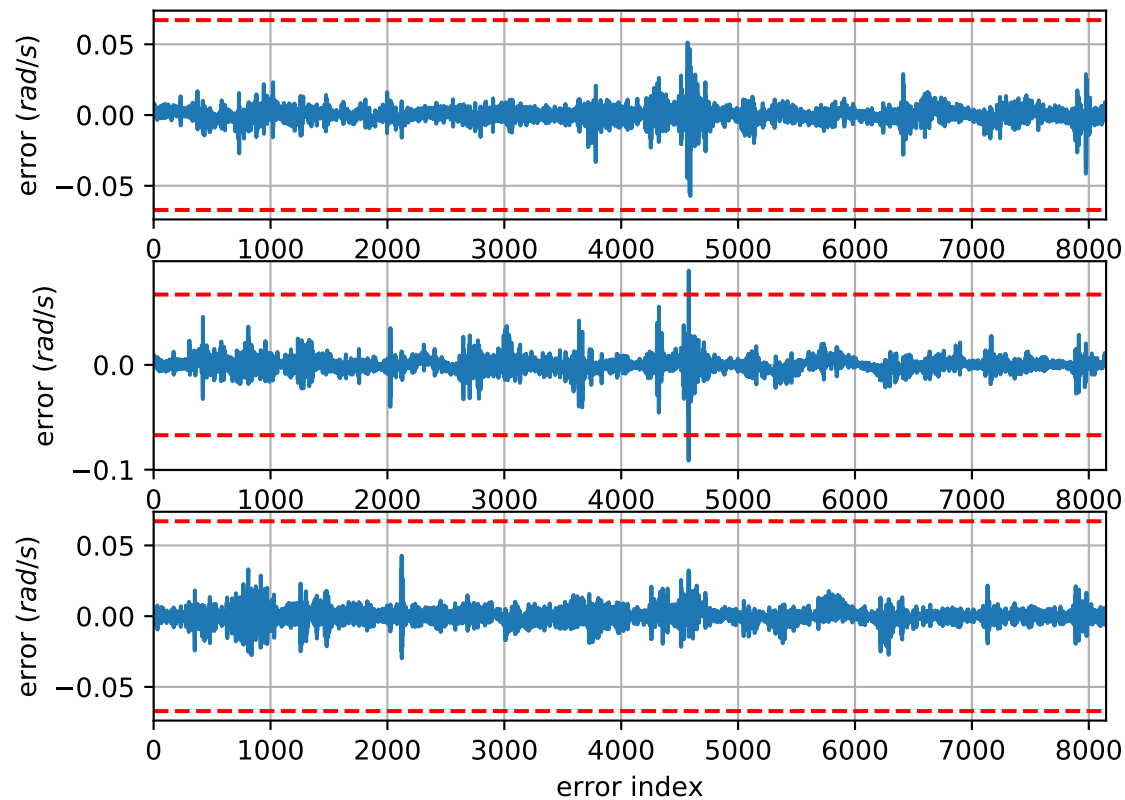
imu0: estimated accelerometer bias (imu frame)



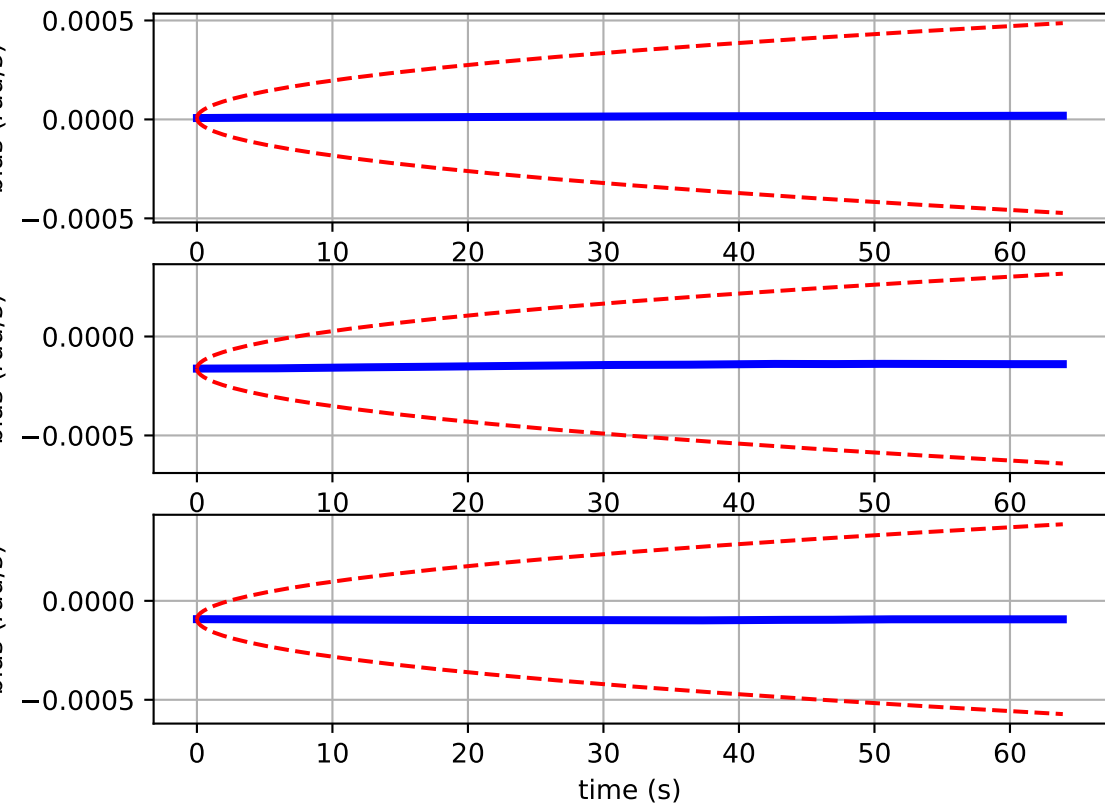
Comparison of predicted and measured angular velocities (body frame)



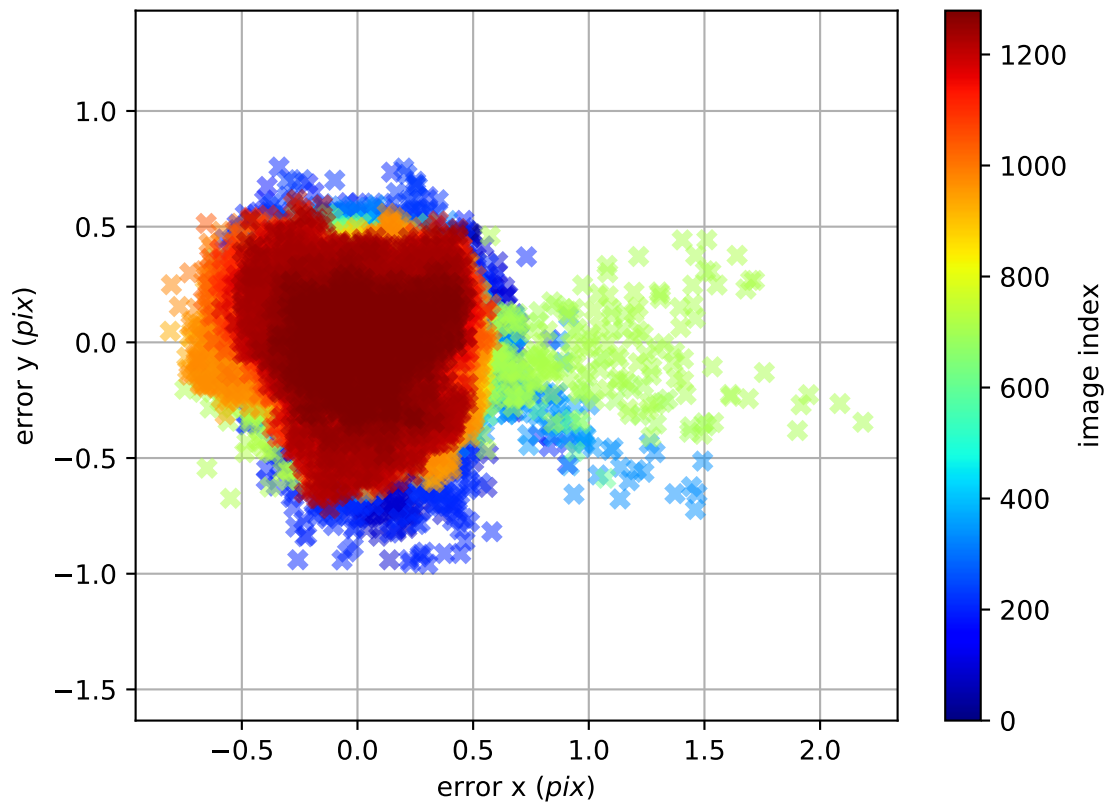
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

